Spinning two-sphere impact

LAMMPS dem06 | Spin-induced friction and angular momentum | 28 October 2026

PASS	4 of 4 runs pass all three official 7% checks at 1.3 ms.

Official rebound checks at 1.3 ms

Run	vx (m/s)	Err. (%)	vz (m/s)	Err. (%)	omega_y	Err. (%)	7%
CPU dt	0.12706942	5.89118	0.09986565	0.13435	19.6839496	0.88646	PASS
GPU dt	0.12706942	5.89118	0.09986565	0.13435	19.6839496	0.88646	PASS
CPU dt/2	0.12705825	5.88188	0.10004190	0.04190	19.6842301	0.88504	PASS
GPU dt/2	0.12705825	5.88188	0.10004190	0.04190	19.6842301	0.88504	PASS

Targets: vx = 0.12 m/s; vz = 0.10 m/s; omega_y = 19.86 rad/s. The official spin_impact checker tests all three relative errors separately against 7%, at step 1300 (second segment end). The 1.8 ms endpoint and golden-trajectory comparisons are separate diagnostics.

Official inputs and direct contact mapping

Quantity	Official input / FunDEM configuration
Two free spheres	Small: R = 0.1 m, density 2700 kg/m^3. Large: R = 0.5 m, density 2.7e6 kg/m^3; mass ratio 125000. Both partners integrate freely.
Initial state	Small center (0, 0, 0.60001) m, v = (0, 0, -0.2) m/s, omega = (0, 22.86, 0) rad/s. Large center, velocity and spin are zero.
Normal / tangential	Hooke kn = 1.2e8 N/m; linear_history kt = 3.4e7 N/m; e = 0.5; sliding mu = 0.40.
Explicit damping	cn = 2 beta sqrt(m* kn) = 15874.46691 N s/m; beta = -ln(e)/sqrt(pi^2 + ln(e)^2); full two-body reduced mass m*. ct = cr = ctorsion = 0.
Other settings	No gravity, wall or global damping. Rolling/torsional stiffness and friction = 0. Elastic material fields are inactive under explicit linear stiffness.
Time / full output	dt = 1 / 0.5 us; 1800 / 3600 steps; same 1.8 ms endpoint. Native frames and CSV are saved every step.

Analytic impulse and spin transfer

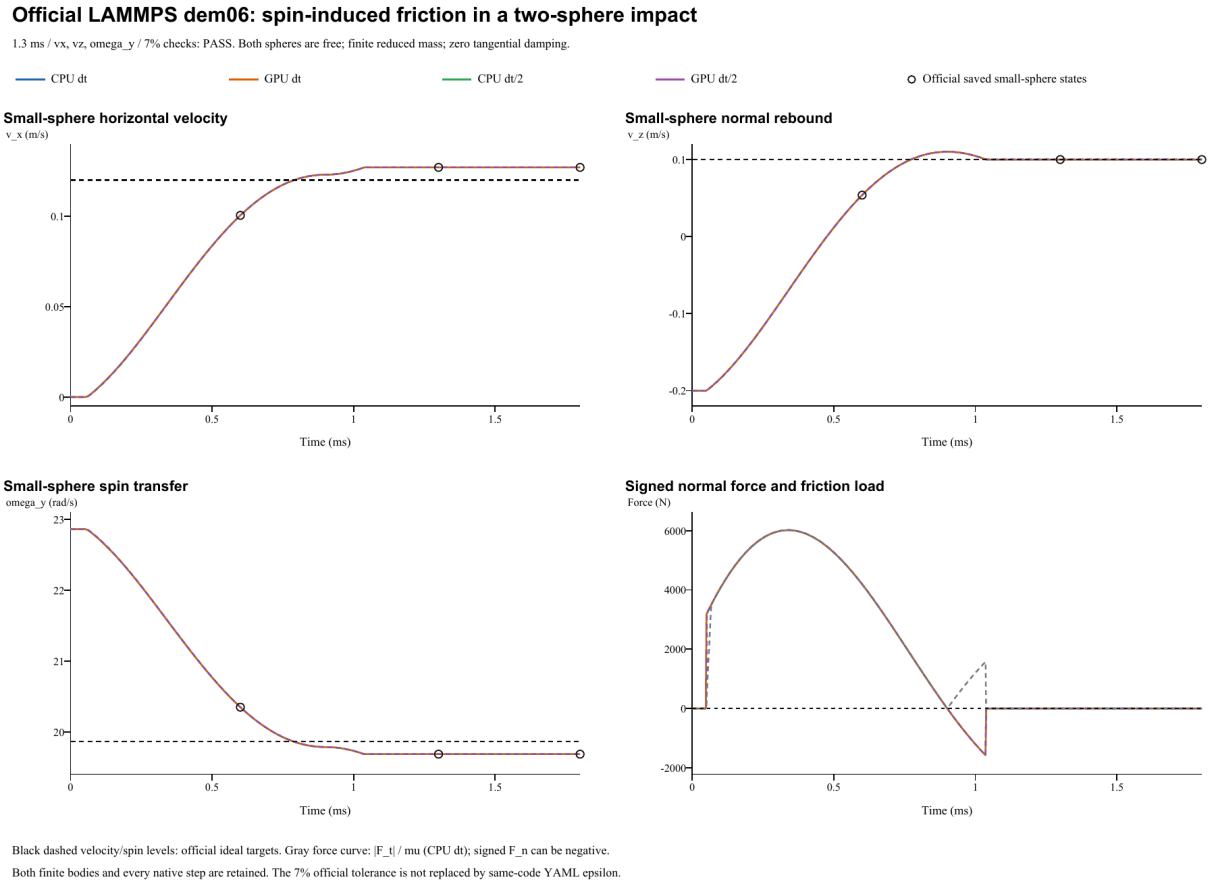
Jn = m_small (1+e) vin; Jt = mu Jn.
vx = mu (1+e) vin = 0.12 m/s; vz = e vin = 0.10 m/s.
omega_y = 22.86 - (5/2) mu (1+e) vin / R_small = 19.86 rad/s.

The analytic checker uses a heavy-partner, gross-sliding approximation. The actual large sphere has finite mass and inertia. Explicit cn preserves official ct = 0; automatic Hooke/tangential damping is not tested.

Reference provenance

Official YAML and golden states | Analytic checker. Commit f9720ce2d176d615495d1d6c31349b1f71be1c73; YAML blob d1cc7b040decc016ec05e95b5e0dc2e283406434. LAMMPS was not rerun locally. YAML epsilon 1e-10 is not a universal cross-code tolerance.

Native histories and retained evidence



Force, impulse and momentum evidence

Base CPU: signed / absolute normal impulse = 3.39143829 / 3.61950266 N s; absolute load is 6.7247% larger. Tangential x impulse = 1.43705473 N s; 18 early uncapped samples reduce the realized friction impulse. Across all four runs, slip stays negative; minimum magnitude = 1.841472 m/s.

The measured small-sphere angular impulse is -0.143681134 N m s. Maximum total linear / angular momentum drifts across four runs are 1.88e-14 kg m/s / 1.89e-14 kg m^2/s, including both bodies and orbital momentum.

CPU/GPU maximum velocity / spin differences are 1.17e-14 m/s / 1.56e-13 rad/s. Halving dt changes checked CPU vx, vz and spin by 1.12e-05 m/s, 0.000176 m/s and 0.00028 rad/s. The approximately 5.9% vx gap to the simplified analytic target remains.

The Coulomb cap uses abs(Fn), while signed Fn sets momentum. Both free particles are included in impulse and total momentum checks. Both codes use pair-force torque arms R_i - delta/2. For surface slip, LAMMPS uses nominal radii while FunDEM uses the shared point; history transport also differs. No target, radius or inertia is adjusted.

Complete native results are retained

CPU dt: run.json | GPU dt: run.json | CPU dt/2: run.json | GPU dt/2: run.json

1801 native frames per base-step run; 3601 per half-step run. All 10,804 frames, contacts, input projects, every-step CSVs, audits and CPU/CUDA backend logs are retained.

Run manifest | Analysis JSON | Checkpoint CSV | Contact audit | Full native archive ZIP

Execution and scope

Real SimulationSession runs use linked FunDEM 0.4.3 CPU/CUDA native libraries without fitting or solver changes. GUI executable byte identity was not established. This case checks spin-induced friction and two-body momentum transfer; torsional resistance about the contact normal is a separate mechanism.